Machine Learning with Applications 23 (2026) 100821

Contents lists available at ScienceDirect

Machine Learning with Applications

journal homepage: www.elsevier.com/locate/mlwa

SAFE AI metrics: An integrated approach

Paolo Giudici *, Vasily Kolesnikov

Department of Economics and Management, University of Pavia, Via San Felice al Monastero 5, 27100 Pavia, Italy

ARTICLE INFO

Keywords:
AI governance
AI risk management
Responsible AI
SAFE AI
Accuracy
Explainability
Security

ABSTRACT

We contribute to the field of AI governance with the development of a unified compliance metric that integrates three key dimensions of SAFE Artificial Intelligence: Security, Accuracy, and Explainability. While these aspects are typically assessed in isolation, the proposed approach integrates them into a single and interpretable metric, grounded in a consistent mathematical structure. To develop an integrated framework, the outputs of machine learning models are evaluated under three risk dimensions, that correspond to different input data perturbations: data removal (for accuracy); data poisoning (for security); and feature removal (for explainability). The experimentation of the methodology on both real and simulated datasets shows that the integrated metric improves compliance monitoring and enables a consistent evaluation of AI risks.

1. Literature review and motivation

Machine learning (ML) is rapidly advancing the capabilities of Artificial Intelligence (AI) across a wide range of human activities. Unlike conventional software systems, AI systems not only transform inputs into outputs but can also influence and reshape their environments. The dynamic and adaptive nature of AI introduces new categories of risk that affect individuals, organizations, and the broader structures of society. To address these emerging risks, regulators, policymakers, and standards organizations have developed AI governance regulations, guidelines and standards, such as European Commission (2020, 2023), International Organization for Standardization and International Electrotechnical Commission (2023), National Institute for Standards and Technology (2023) and United Nations High-Level Advisory Body on Artificial Intelligence (2024).

These initiatives emphasize the importance of systematically assessing and mitigating the risks associated with AI. A key aspect for the governance of AI, and of its risk management, is the establishment of consistent and interpretable metrics, that can guide monitoring and ensure compliance. However, despite increasing attention to AI risk management, a standardized set of risk metrics for AI systems is still lacking, in both the academic literature and in the professional practice, see e.g. Bernardelli and Giudici (2025).

In response to this challenge, Babaei et al. (2025) summarized the existing governance guidelines and introduced four measurable principles under the acronym S.A.F.E:

- **Sustainability**: This principle requires that AI systems are robust and resilient to adversarial attacks, such as cyber attacks or data poisoning, and is commonly measured by means of red teaming procedures, see e.g. Ferrer et al. (2024) and Zhao et al. (2022) for a review.
- **Accuracy**: This principle require that AI systems make correct predictions, and it is often quantified using metrics such as Root Mean Squared Error (RMSE) for continuous outcomes and the Area Under the Curve (AUC) for binary classifications, see e.g. Gneiting and Katzfuss (2014) and Terven et al. (2025).
- **Fairness**: This principles requires that AI systems behave similarly for different population groups, and is commonly evaluated using parity-based metrics such as demographic parity or equalized odds, see e.g. Hurlin et al. (2024)
- **Explainability**: This principle requires that the output of AI systems should be understood in its drivers, to allow its monitoring and control, and is mostly measured by post-hoc techniques such as Shapley Values and LIME, see e.g. Lundberg and Lee (2017), Molnar (2019) and Ribeiro et al. (2016).

Although widely used metrics and procedures such as red teaming, AUC, RMSE, Shapley Values and fairness parity measures address individual aspects of the S.A.F.E-AI principles, they are generally applied in isolation and lack internal consistency. This fragmentation limits their ability to provide a comprehensive assessment of AI risk.

This limitation has been acknowledged in some recent papers, such as Liang et al. (2022) and Nguyen and Do (2025). In Nguyen and Do (2025) the authors introduce RAISE (Responsible AI Scoring and Evaluation): "a unified framework that systematically quantifies model performance across the foundational and often competing dimensions

* Corresponding author.
E-mail addresses: paolo.giudici@unipv.it (P. Giudici), vasily.kolesnikov01@universitadipavia.it (V. Kolesnikov).

https://doi.org/10.1016/j.mlwa.2025.100821
Received 20 July 2025; Received in revised form 28 November 2025; Accepted 9 December 2025
Available online 15 December 2025
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of explainability, fairness, robustness, and sustainability". They propose to standardize and integrate the existing individual metrics for the latter four dimension, leading to a unified responsibility score for tabular data that can be put in trade-off with accuracy. In Liang et al. (2022) the authors propose a similar approach to evaluate Large Language models, considering a multi-metric framework that integrates seven dimensions: accuracy, calibration, robustness, fairness, bias, toxicity and efficiency. Both papers employ existing individual metrics, standardize them to a common metric and, then, integrate them in a single measurement.

The advantage of a unified metric approach to responsible AI and AI risk management is a transparent manner to evaluate the output of AI applications, which leads to a single, interpretable, and task-agnostic S.A.F.E-AI score. The resulting framework improves the ability of different stakeholders, such as AI providers, users and regulators, to monitor and manage AI-related risks.

We follow a similar approach: consider a multi-metric framework that integrates four dimensions: sustainability, accuracy, fairness and explainability into a single metric. And, similarly as in Nguyen and Do (2025), we consider tabular data as a reference case. However, differently from Nguyen and Do (2025) we do not consider different metrics but, rather, measure all dimensions with a common metric, derived from a common mathematical basis: the Lorenz curve and its relationship to the Receiver Operating Characteristic (ROC) curve (see, e.g. Giudici & Raffinetti, 2024; Koshevoy & Mosler, 1996; Lorenz, 1905).

The advantage of our proposal is a precise determination of the relative importance of each dimension, as they are all measured with the same metric. A possible disadvantage is that we are not using existing metrics but, rather, new ones, which need to be understood, especially in comparison with the standard ones.

2. Proposal

The RGB framework (Babaei et al., 2025) consists of three fundamental mathematical constructs: the Lorenz curve, the dual Lorenz curve, and the concordance curve. These curves form a unified basis for defining the proposed risk and compliance metrics.

Consider two statistical variables, Y^* and Y^{**}, which may be continuous, ordinal or binary. The Y^* values can be used to build the Lorenz curve, L_{Y^*}, arranging the Y^* values in a non-decreasing order. For $i = 1, \dots, n$, the Lorenz curve can be constructed by the pairs:

$$\left(\frac{i}{n}, \frac{1}{n\bar{y}^*} \sum_{j=1}^{i} y^*_{r^*_j} \right), \tag{1}$$

where r^*_j indicates the non-decreasing ranks of Y^* and $\bar{y}^*$ denotes the mean of Y^*.

In contrast, the dual Lorenz curve is created by ordering the same Y^* values in a non-increasing order and is defined by pairs:

$$\left(\frac{i}{n}, \frac{1}{n\bar{y}^*} \sum_{j=1}^{i} y^*_{r^*_{n+1-j}} \right), \tag{2}$$

where r^*_{n+1-j} indicates the non-increasing ranks of Y^*.

With the same principle, we can order the Y^{**} values. Let r^{**}_i, for $i = 1, \dots, n$, indicate the non-decreasing ranks of Y^{**}. A concordance curve C can be constructed by ordering the Y^* values not in terms of their own ranks, but with respect to the ranks of Y^{**}. For $i = 1, \dots, n$, the concordance curve is defined by the pairs:

$$\left(\frac{i}{n}, \frac{1}{n\bar{y}^*} \sum_{j=1}^{i} y^*_{r^{**}_j} \right), \tag{3}$$

where r^{**}_i indicates the non-decreasing ranks of Y^{**}.

To illustrate these concepts, Fig. 1 presents the concordance curve C along with the Lorenz curve and the dual Lorenz curve for the binary classification problem.

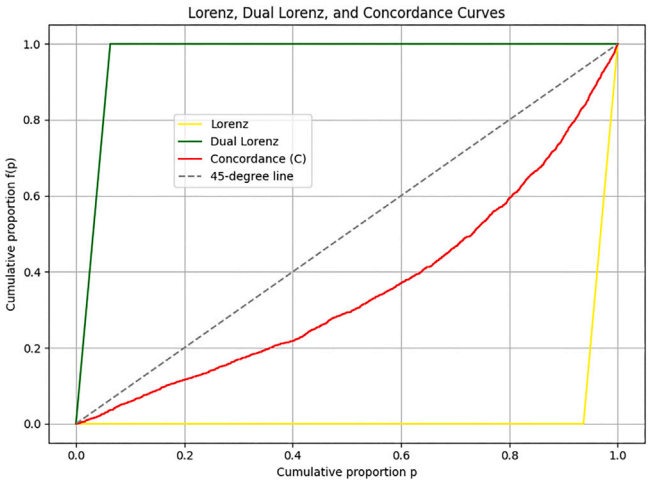


Fig. 1. The Lorenz L_{Y^*} and L'_{Y^*} dual Lorenz curves, the concordance curve C, and the 45-degree line, where p (on the x-axis) and $f(p)$ (on the y-axis) represent the cumulative values of the x and y coordinates of the L_{Y^*}, L'_{Y^*}, and C curves.

Importantly, the concordance curve coincides exactly with the Lorenz curve when the rankings of Y^* and Y^{**} are identical, and it matches the dual Lorenz curve when the rankings are perfectly reversed. More generally, the difference between the concordance and Lorenz curves at any point reflects the degree to which the rank orderings of the two variables differ, providing a measure of their concordance or divergence in terms of their distributions.

2.1. Rank Graduation Accuracy (RGA)

Accuracy reflects how closely the output of an AI system aligns with the expected or true values. For instance, in the NIST Risk Management Framework (National Institute for Standards and Technology, 2023), accuracy is described as the degree to which results or estimates approximate actual or accepted reference values.

In both statistics and machine learning, accuracy is traditionally measured using well-established metrics. For continuous outcomes, the Root Mean Squared Error (RMSE) is commonly used, while for binary classification tasks, the Area Under the Receiver Operating Characteristic Curve (AUROC) serves as a standard performance indicator.

To address the limitations of task-specific accuracy metrics, a more general, response-agnostic measure was proposed—Rank Graduation Accuracy (RGA). An important property is that the RGA measure is equal to the AUROC when applied to binary outcomes. Also, it extends naturally to ordinal and continuous targets, preserving a consistent interpretation across different problem types.

Dividing the area under the concordance curve by its maximum value, we obtain the RGA measure:

$$RGA = \frac{\sum_{i=1}^{n} \left\{ \frac{1}{n\bar{y}} \left(\sum_{j=1}^{i} y_{r_{n+1-j}} - \sum_{j=1}^{i} y_{\hat{r}_j} \right) \right\}}{\sum_{i=1}^{n} \left\{ \frac{1}{n\bar{y}} \left(\sum_{j=1}^{i} y_{r_{n+1-j}} - \sum_{j=1}^{i} y_{r_j} \right) \right\}} = \frac{\sum_{i=1}^{n} i y_{\hat{r}_i} - \sum_{i=1}^{n} i y_{r_{n+1-i}}}{\sum_{i=1}^{n} i y_{r_i} - \sum_{i=1}^{n} i y_{r_{n+1-i}}}, \tag{4}$$

where, for any $i = 1, \dots, n$: r_i are the non-decreasing ranks of the actual response values y_i; $\hat{r}_i$ are the non-decreasing ranks of the predicted values $\hat{y}_i$; r_{n+1-i} are the non-increasing ranks of the actual response values y_i; $\bar{y}$ is the mean value of the actual response values y_i.

It can be shown that the values of the RGA metric lie in the interval $0 \leqslant RGA \leqslant 1$, with $RGA = 1$ for a perfectly concordant model ($C \equiv L_{\hat{Y}}$);

$RGA = 0$ for a perfectly discordant model ($C \equiv L'_{\hat{Y}}$); $RGA = 0.5$ for random predictions (C coincides with the 45-degree line).

We propose a decomposition of the RGA metric into a vector of partial RGA values, each corresponding to a segment of the ranked predictions. This approach goes beyond the use of a single summary score and provides a more detailed view of how different parts of the data contribute to the overall accuracy of the model.

This decomposition is particularly important in the context of the unified compliance score, where we simulate cyber-attacks that selectively damage or degrade parts of the data. In such scenarios, relying solely on the final RGA score is insufficient, as it may hide local weaknesses or failure points. By breaking the RGA into partial contributions, we can assess how specific segments of the model's ranking respond to such disruptions. This allows for a more robust evaluation of the model accuracy under adverse conditions.

More specifically, we remove data proceeding from the highest-ranked values to the lowest-ranked values, following a reversed construction, compared to the usual Lorenz and concordance curves. The dataset is divided into k equal-sized segments, where for each segment $k = 1, \dots, n$, we calculate a partial RGA contribution, denoted RGA_k, using the same RGA formula, but restricted to the kth slice of the data. These partial contributions are normalized by the total area between the dual Lorenz and Lorenz curves to ensure that their cumulative sum equals the full RGA score. The complete RGA value corresponds to the sum of all partial contributions:

$$RGA = \sum_{k=1}^{n} RGA_k. \tag{5}$$

To construct the final vector, we accumulate the partial contributions in reverse order, starting from the full RGA and decrementing it by each slice's contribution :

$$\left[RGA, \quad RGA - RGA_1, \quad RGA - (RGA_1 + RGA_2), \quad \dots, \quad 0\right]. \tag{6}$$

This cumulative vector provides a monotonically decreasing, stepwise approximation of the concordance curve's area, where each step indicates the remaining predictive power when progressively removing contributions from the highest-ranked segments down to the lowest-ranked ones.

Building on the previous procedure, we can define The Area Under the Rank Graduation Accuracy Curve (AURGA) and formally write the formula:

$$AURGA = \int_0^1 RGA(x)dx, \tag{7}$$

where $RGA(x)$ is the piecewise constant (step) function defined by our discrete cumulative vector, with x representing the proportion of the highest-ranked segments considered.

Fig. 2 shows an illustration of how RGA can be derived, for a Random Forest model applied to the California HMDA dataset that will be considered in the application.

2.2. Rank Graduation Robustness (RGR)

Robustness in machine learning generally refers to a model's ability to maintain stable performance under perturbations or unexpected conditions. The robustness literature addresses various dimensions including adversarial attacks, natural corruptions, distribution shifts, and algorithmic stability (Cohen et al., 2019; Hendrycks & Dietterich, 2019; National Institute for Standards and Technology, 2025; Rousseeuw & Leroy, 1987). In this work, we focus on adversarial attacks by measuring how model performance degrades under systematic perturbations to assess stability across different architectures.

Most robustness metrics quantify change in accuracy or rankings after perturbation. The S.A.F.E-AI framework introduces Rank Graduation Robustness (RGR), which, for each feature, compares the ranks of predictions on original versus perturbed input.

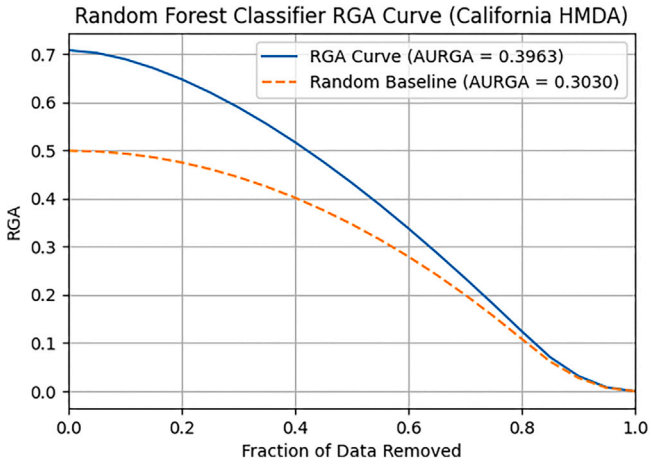


Fig. 2. Rank Graduation Accuracy (RGA) Curve of the Random Forest Classifier on the California Home Mortgage Disclosure Act (HDMA) dataset, in comparison with a random baseline. The Figure shows the superiority of the Random Forest model, up to a 90% data removal.

Dividing the area under the concordance curve by its maximum value, we obtain the RGR measure:

$$RGR = \frac{\sum_{i=1}^{n}\left\{\frac{1}{n\bar{\hat{y}}}\left(\sum_{j=1}^{i}\hat{y}_{r_{n+1-j}} - \sum_{j=1}^{i}\hat{y}_{r_j^p}\right)\right\}}{\sum_{i=1}^{n}\left\{\frac{1}{n\bar{\hat{y}}}\left(\sum_{j=1}^{i}\hat{y}_{r_{n+1-j}} - \sum_{j=1}^{i}\hat{y}_{r_j}\right)\right\}} = \frac{\sum_{i=1}^{n} i\hat{y}_{r_i^p} - \sum_{i=1}^{n} i\hat{y}_{r_{n+1-i}}}{\sum_{i=1}^{n} i\hat{y}_{r_i} - \sum_{i=1}^{n} i\hat{y}_{r_{n+1-i}}}, \tag{8}$$

where, for any $i = 1, \dots, n$: r_i are the non-decreasing ranks of the predicted values $\hat{y}_i$ computed on data without perturbations; r_i^p are the non-decreasing ranks associated with the predicted values computed on perturbed data; r_{n+1-i} are the non-increasing ranks of the predicted values $\hat{y}_i$ computed on data without perturbations; $\bar{\hat{y}}$ is the mean value of the predicted values $\hat{y}_i$ computed on data without perturbations.

This metric has the same normalization properties as RGA. The values of the RGR metric lie in the interval $0 \leqslant RGR \leqslant 1$, with $RGR = 1$ for a full robust model ($C \equiv L_{\hat{Y}}$); $RGR = 0$ for a full perturbed model ($C \equiv L'_{\hat{Y}}$); $RGR = 0.5$ if perturbations lead to a random model (C coincides with the 45-degree line).

We evaluate robustness through two perturbation strategies that test different aspects of model stability. A percentile swapping method introduces controlled, distribution-preserving perturbations by swapping values between the lower (pth percentile) and upper ($(1-p)$th percentile) tails of each variable. Whereas a Gaussian noise method adds random noise $\mathcal{N}(0, (\alpha \cdot \sigma_{\text{data}})^2)$ across all features simultaneously, simulating realistic measurement errors and sensor noise commonly encountered in real-world applications.

Building on the RGR definition, we propose to calculate the Area Under the Robustness Curve (AURGR) as a single summary metric of model robustness to data perturbations. First, we generate the robustness curve by plotting the RGR score against the perturbation intensity. Since RGR starts at the value 1 when there is no perturbation and then monotonically decreases — typically falling below the random baseline value of 0.5 as perturbation increases — the curve captures how rapidly model performance degrades under increasing perturbations. To ensure comparability across different perturbation methods, we normalize all perturbation intensities to a common scale. We evaluate RGR at raw perturbation intensities ranging from 0% to 50% (where 50% represents complete tail swapping for the percentile method and the multiplier of the data standard deviation for Gaussian noise), then normalize these intensities by dividing by the maximum value to obtain

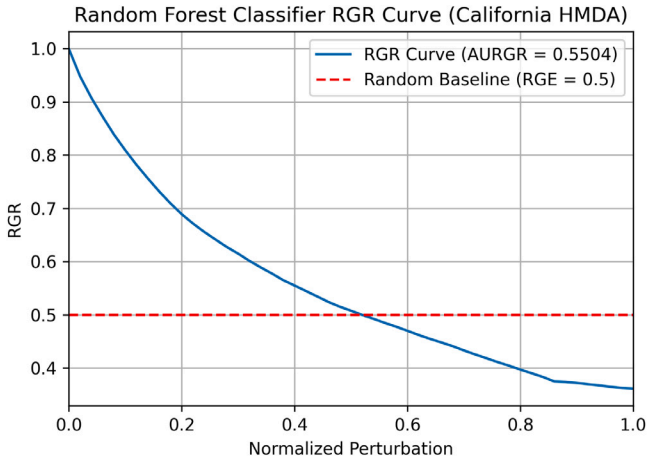


Fig. 3. Rank Graduation Robustness (RGR) Curve of the Random Forest Classifier on the California HMDA dataset (Percentile Swap Perturbation), in comparison with a random baseline. The Figure shows the superiority of the Random Forest model, up to a 50% data perturbation.

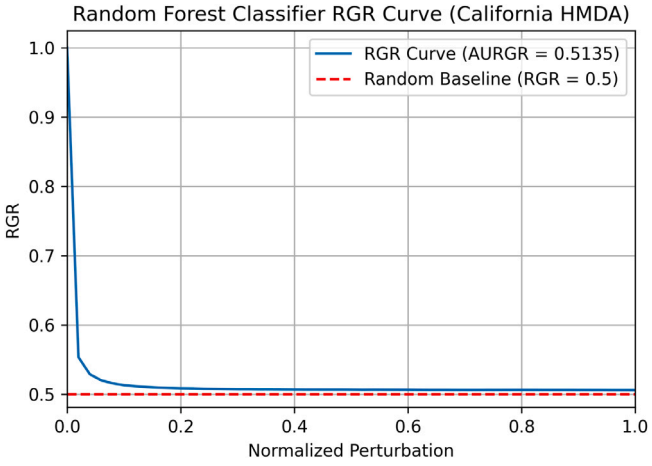


Fig. 4. RGR Curve of the Random Forest Classifier on the California HMDA dataset (Gaussian Noise Perturbation, in comparison with a random baseline. The Figure shows a clear superiority of the model, up to a 20% data perturbation.

$p \in [0,1]$. We then compute

$$\text{AURGR} = \int_0^1 \text{RGR}(p)\,\mathrm{d}p, \tag{9}$$

where $p = t/p_{\max}$ is the normalized perturbation intensity in $[0,1]$, with t being the raw perturbation intensity and $p_{\max}$ be the maximum value. This normalization ensures that AURGR values are directly comparable across perturbation methods, regardless of their natural intensity scales.

Figs. 3 and 4 illustrate the derivation of RGR for a Random Forest model using percentile swap and Gaussian noise perturbations, respectively, on the California HMDA dataset. The percentile swap case shows no initial sharp drop because perturbations are localized to extreme values, whereas Gaussian noise affects all observations simultaneously, immediately degrading model performance.

2.3. Rank Graduation Explainability (RGE)

As machine learning systems have grown in complexity, the demand for transparency and human-understandable reasoning, known as Explainable AI (XAI), has become significant for building trust, ensuring safety and enabling meaningful oversight. Traditional "white-box" models such as linear and logistic regression are relatively easy to interpret, using clear mathematical relationships and well-understood statistical methods. In contrast, modern high-capacity models, such as random forests and deep neural networks, are often seen as 'black boxes' because their complex internal structure make it difficult to understand how specific inputs lead to specific outputs (Doshi-Velez & Kim, 2017).

Explainability is usually defined as the ability to represent how an AI system operates in a form that is understandable to humans (National Institute for Standards and Technology, 2023). This is vital for enabling user monitoring, intervention, and correction of AI system behavior.

Among the several XAI methods available, the most popular is based on Shapley values (Guidotti et al., 2018). However, Shapley-based methods are computationally demanding and become impractical when dealing with a large number of input variables. The Rank Graduation Explainability metric (RGE) proposed in Babaei et al. (2025) overcomes this drawback.

The RGE metric is obtained dividing the area under the concordance curve by its maximum value, and taking its complement:

$$\begin{aligned} RGE &= 1 - \frac{\sum_{i=1}^{n}\left\{\frac{1}{n\bar{\hat{y}}}\left(\sum_{j=1}^{i}\hat{y}_{r_{n+1-j}} - \sum_{j=1}^{i}\hat{y}_{r_j^{-X_k}}\right)\right\}}{\sum_{i=1}^{n}\left\{\frac{1}{n\bar{\hat{y}}}\left(\sum_{j=1}^{i}\hat{y}_{r_{n+1-j}} - \sum_{j=1}^{i}\hat{y}_{r_j}\right)\right\}} \\ &= 1 - \underbrace{\frac{\sum_{i=1}^{n} i\hat{y}_{r_j^{-X_k}} - \sum_{i=1}^{n} i\hat{y}_{r_{n+1-i}}}{\sum_{i=1}^{n} i\hat{y}_{r_i} - \sum_{i=1}^{n} i\hat{y}_{r_{n+1-i}}}}_{\widetilde{RGE}}, \end{aligned} \tag{10}$$

where, for any $i = 1,\ldots,n$: r_i are the non-decreasing ranks of the predicted values $\hat{y}_i$ provided by the full model; $r_i^{-X_k}$ are the non-decreasing ranks associated with the predicted values provided by the model without the kth predictor (with $k = 1,\ldots,K$); r_{n+1-i} are the non-increasing ranks of the predicted values computed on the full model; $\bar{\hat{y}}$ is the mean value of the full model predicted values $\hat{y}_i$.

The values of the RGE metric lie in the interval $0 \leqslant RGE \leqslant 1$: $RGE = 1$ if the kth predictor provides the maximum explanation ($C \equiv L'_{\hat{Y}}$); $RGE = 0$ if the kth predictor does not contribute to the explanation of the response ($C \equiv L_{\hat{Y}}$); $RGE = 0.5$ if the model without the kth predictor corresponds to the random model (C coincides with the 45-degree line).

Based on the RGE framework, we propose the Area Under the Explainability Curve (AURGE) as a single summary metric of model explainability. We can define it as:

$$\text{AURGE} = \int_0^1 \text{RGE}(p)\,\mathrm{d}p, \tag{11}$$

where p denotes the fraction of features excluded. To make the explainability curve decrease as we remove features, we will use a reversed RGE definition:

$$\begin{aligned} RGE^* &= \frac{\sum_{i=1}^{n}\left\{\frac{1}{n\bar{\hat{y}}}\left(\sum_{j=1}^{i}\hat{y}_{r_{n+1-j}} - \sum_{j=1}^{i}\hat{y}_{r_j^{-X_k}}\right)\right\}}{\sum_{i=1}^{n}\left\{\frac{1}{n\bar{\hat{y}}}\left(\sum_{j=1}^{i}\hat{y}_{r_{n+1-j}} - \sum_{j=1}^{i}\hat{y}_{r_j}\right)\right\}} \\ &= \underbrace{\frac{\sum_{i=1}^{n} i\hat{y}_{r_j^{-X_k}} - \sum_{i=1}^{n} i\hat{y}_{r_{n+1-i}}}{\sum_{i=1}^{n} i\hat{y}_{r_i} - \sum_{i=1}^{n} i\hat{y}_{r_{n+1-i}}}}_{\widetilde{RGE}}, \end{aligned} \tag{12}$$

We flip the original RGE so that at $p = 0$ (no features removed) it starts at 1, and at $p = 1$ (all features removed) it ends at the random-baseline value of 0.5. That way, the plotted RGE curve is monotonically decreasing over p.

Fig. 5 shows an illustration of how RGE can be derived using the same Random Forest model and California HMDA dataset as in the previous examples.

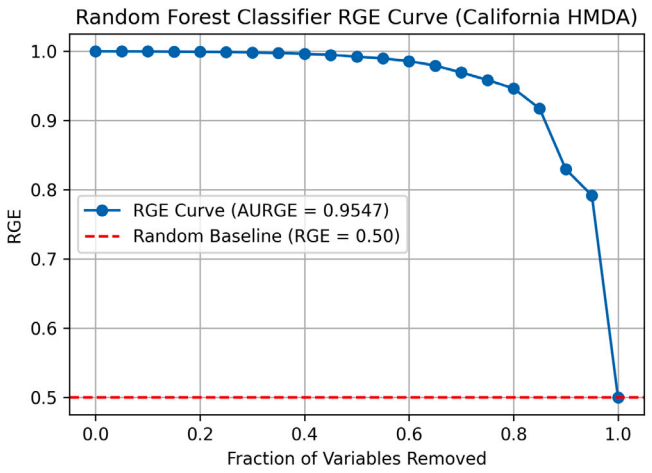


Fig. 5. Rank Graduation Explainability (RGE) Curve of the Random Forest Classifier for the California HMDA dataset. The Figure shows that about 60% of the variables can be removed.

2.4. Integration

We now describe how the three metrics can be integrated in a single measurement. We will compute a normalized volume under a three-dimensional surface defined by RGA, RGE and RGR metric vectors. Given the three RGA, RGE and RGR vectors of equal length, n, (corresponding to the number of features), we construct a 3D grid where: RGA captures how data quality degrades as input data is removed, RGE reflects how model explainability is affected when features are removed, and RGR shows the model's robustness against increasingly additional noise.

For each triplet of indices (i, j, k) where $i, j, k \in \{1, 2, \ldots, n\}$, we apply a mean aggregation function to the corresponding metric values $(RGA[i], RGE[j], RGR[k])$. We will use three alternative mean calculations:

$$\text{Arithmetic mean:} \quad M(i,j,k) = \frac{RGA[i] + RGE[j] + RGR[k]}{3} \tag{13}$$

$$\text{Geometric mean (Cube):} \quad M(i,j,k) = \sqrt[3]{RGA[i] \times RGE[j] \times RGR[k]} \tag{14}$$

$$\text{Quadratic mean (Root mean square):} \quad M(i,j,k) = \sqrt{\frac{RGA[i]^2 + RGE[j]^2 + RGR[k]^2}{3}} \tag{15}$$

This creates an $n \times n \times n$ tensor $\mathbf{M}$ representing a discrete approximation of a continuous 3D surface:

$$\mathbf{M} = \{M(i,j,k) : i,j,k \in \{1,2,\ldots,n\}\} \tag{16}$$

The final normalized volume is computed as:

$$\mathrm{V} = \frac{1}{n^3} \sum_{i=1}^{n} \sum_{j=1}^{n} \sum_{k=1}^{n} M(i,j,k) \tag{17}$$

This normalized volume represents the average "height" of the 3D surface across the entire cyber attack scenario space, providing a unified metric that captures the joint behavior of accuracy, explainability, and robustness under different conditions. We will term this unified measure the **Compliance Score**, as it assesses how well a model complies with multiple critical requirements simultaneously: performance standards (accuracy), transparency regulations (explainability) and security requirements (robustness). This compliance assessment provides a single metric for evaluating model readiness across the comprehensive standards expected in regulated production environments. We will visualize the distribution of compliance scores across different models using histogram plots to facilitate comparative analysis and identify performance patterns.

For comparison, we will compare our aggregation methods with TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution) (Hwang & Yoon, 1981), a multi-criteria decision analysis method that identifies the best alternative by minimizing distance from the ideal solution and maximizing distance from the negative-ideal solution across all metrics, providing an alternative aggregation perspective.

We remark that our approach is agnostic with respect to risk preferences—we do not assign weights to the three dimensions but rather take simple averages across all perturbation combinations. While it is beyond our scope, we will briefly demonstrate one example of weight assignment, showing that this may be a promising direction for future research developments.

3. Data

To illustrate our methodology, we utilize data from the Home Mortgage Disclosure Act (HMDA), the most comprehensive publicly available dataset on the U.S. mortgage market. Enacted in 1975 and implemented through Regulation C, HMDA requires various financial institutions to report detailed information on individual mortgage applications. This data serves multiple critical functions: HMDA helps assess whether lenders are meeting community credit needs, helps policy makers develop equitable housing initiatives, and highlights potential patterns of discriminatory lending. To ensure the borrower's privacy, the publicly released data is anonymized. We examine two different subsets of HMDA data for the loan applications filed in 2017: the New York dataset which consists of 44 6902 loans, and the California dataset containing 1 714 459 loans.

After filtering out irrelevant and non-informative features and reducing the number of categories in some variables, we focus on 21 features for further analysis, as described in Table 1.

For comparison we also apply the proposal to a synthetic dataset, which contains 3000 employee records with eight features including salary, age, education, job tenure, previous experience, minority status and gender. The binary target variable follows an XOR (exclusive OR) pattern based on age and salary, where the positive class is assigned when exactly one of two conditions holds: either an employee is young (below median age) with high salary (above median), or older (above median age) with low salary (below median). This XOR relationship creates a non-linear decision boundary that cannot be captured by any single hyperplane, as the two target classes occupy diagonally opposite quadrants in the age-salary feature space. Consequently, linear models such as logistic regression are fundamentally incapable of learning this pattern without feature transformations or interactions. This design makes this dataset suitable for detecting meaningful variations in our metrics that might be obscured in datasets where all models perform similarly.

4. Empirical findings

This section presents a comparative analysis of models performance across our datasets using AURGA, AURGR, and AURGE metrics. We evaluate AURGR under two perturbation strategies (Swapping and Gaussian noise), compare RGE results with Shapley values and demonstrate the impact of weight assignments on model rankings.

A diverse set of machine learning models was applied. These models are used as tools to generate predictions upon which the new scoring methodology is evaluated. The selection includes models from different algorithmic families: logistic, tree-based, ensemble and neural network approaches, ensuring a well-rounded and representative set of models.

Table 1
Summary of HMDA variables.

Variable	Description	Values
Response variable		
action_taken	Loan approval outcome	Application denied, Loan originated
Binary variables		
loan_type	Categories of the loan	Conventional, FHA-insured
applicant_sex	Applicant gender	Male, Female
lien_status	Collateral priority	Secured by a first lien, Secured by a subordinate lien
applicant_ethnicity	Applicant ethnicity	Hispanic or Latino, Not Hispanic or Latino
Categorical variables		
loan_purpose	Purpose of the loan	Home purchase, Home improvement, Refinancing
co_applicant_ethnicity	Co-applicant ethnicity	Hispanic or Latino, Not Hispanic or Latino, No co-applicant
applicant_race_1	Applicant race	White, Black or African American, Asian(for California)
co_applicant_race_1	Co-applicant race	White, Black or African American, Asian(for California), No co-applicant
co_applicant_sex	Co-applicant gender	Male, Female, No co-applicant
Continuous variables		
loan_amount_000s	Loan Amount in thousands of dollars	Continuous
applicant_income_000s	Gross Annual Income in thousands of dollars	Continuous
county_code	Three-digit FIPS county identifier	Continuous
census_tract_number	Census tract number	Continuous
msamd	Metropolitan Statistical Area	Continuous
population	Total population in tract	Continuous
minority_population	Percentage of minority population to total population for tract	Continuous
hud_median_family_income	Median family income fro the tract	Continuous
tract_to_msamd_income	Percentage of tract median family income compared to MSA/MD median family income	Continuous
number_of_owner_occupied_units	Number of dwellings, including individual condominiums, that are lived in by the owner	Continuous
number_of_1_to_4_family_units	Dwellings that are built to house fewer than 5 families	Continuous

Table 2
AURGA comparison between datasets for each model.

Model	New York HMDA	California HMDA	Synthetic employee
LR	0.3837	0.3709	0.4818
RF	0.4121	0.3963	0.9999
XGB	0.4133	0.3995	0.9999
SEM	0.4137	0.3997	0.9999
VEM	0.4138	0.3993	0.9999
NN	0.4100	0.3960	–
Random	0.3024	0.3030	0.4978

4.1. AURGA results

We first compare models in terms of AURGA. Table 2 contains results for all considered data and Fig. 6 visualize the RGA curves, for the California HMDA data (as representative of the different model behavior). Note that the Neural Network model was not evaluated on the Employee dataset due to insufficient data for reliable training.

Based on the comparison, here are the key conclusions:

- On real-world data all models outperform the random baseline as expected.
- On HMDA datasets, ensemble methods (Voting and Stacking) slightly outperform individual models on New York dataset but show nearly identical performance with XGBoost on California dataset, with all algorithms demonstrating competitive performance within a narrow 0.01–0.03 range.

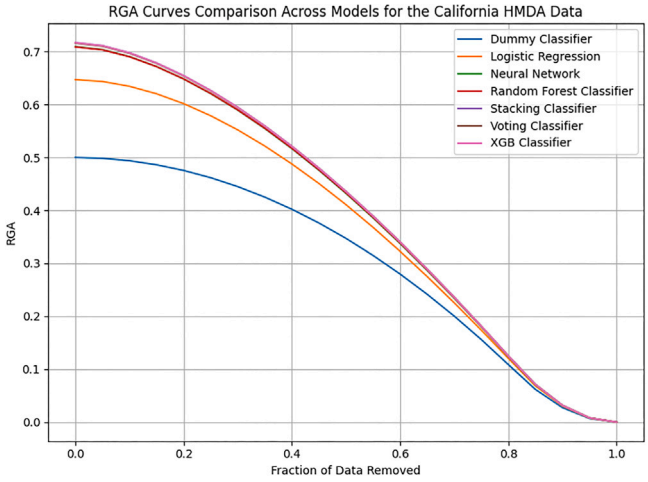


Fig. 6. Rank Graduation Accuracy (RGA) Curves of all models for the California HMDA dataset. The curves indicates that Stacking, Voting and Gradient Boosting Classifier (XGB) classifiers are the best models.

- The Employee dataset exposes fundamental model limitations: Logistic Regression achieves only 0.4818 (worse than random performance), while all other models reach near-perfect scores (0.9999), demonstrating their ability to capture the XOR pattern.

Table 3
AURGR comparison (Swapping) between datasets for each model.

Model	New York HMDA	California HMDA	Synthetic employee
LR	0.4484	0.4619	0.3121
RF	0.5181	0.5504	0.6778
XGB	0.5372	0.5557	0.6665
SEM	0.5279	0.5580	0.6779
VEM	0.5289	0.5531	0.6780
NN	0.5242	0.5481	–
Random	1.0	1.0	1.0

Table 4
AURGR comparison (Gaussian noise) between datasets for each model.

Model	New York HMDA	California HMDA	Synthetic employee
LR	0.9807	0.9668	0.9815
RF	0.5070	0.5135	0.5133
XGB	0.5082	0.5104	0.5167
SEM	0.5096	0.5114	0.5180
VEM	0.5078	0.5123	0.5166
NN	0.7608	0.8152	–
Random	1.0	1.0	1.0

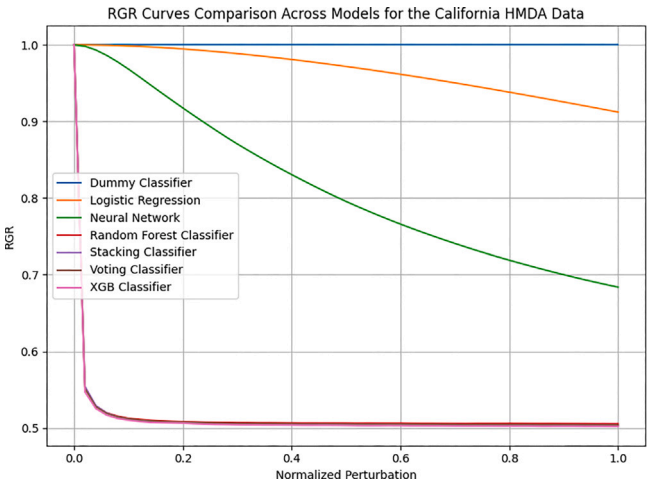


Fig. 8. Rank Graduation Robustness (RGR) Curves of all models for the California HMDA dataset (Gaussian Noise). The curves indicate that the Logistic regression and the Neural network are the most robust models.

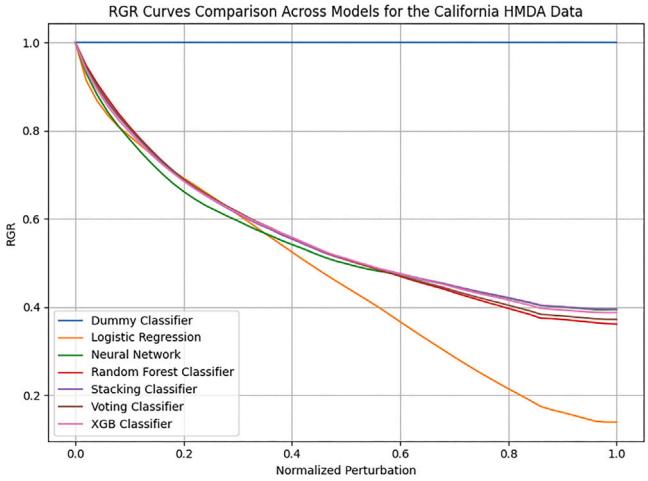


Fig. 7. Rank Graduation Robustness (RGR) Curves of all models for the California HMDA dataset (Swapping perturbation.) The curves indicate that the Random Forest, Gradient Boosting, Voting and Stacking are the most robust models.

- Logistic Regression is the weakest performer across all datasets, with the Employee data magnifying this weakness—the performance gap between LR and non-linear models is 0.51 on the Employee data with only 0.03 gap on HMDA datasets.

4.2. AURGR results

We now compare models in terms of AURGR across different perturbation methods. Table 3 contains results for all considered datasets using the Swapping perturbation method, while Table 4 presents results for Gaussian noise.

Figs. 7 and 8 visualize the RGR curves using our perturbation methods for the California HMDA data.

Based on the AURGR comparison tables and graphs, here are the key conclusions:

- Almost all models achieve AURGR scores significantly below the Random baseline (1.0), which by construction is perfectly robust as it remains unaffected by input perturbations. The random baseline is just a benchmark, and not a model, as it does not depend on the input data.
- The choice of perturbation method reveals drastically different robustness patterns. Under swapping perturbations, tree-based methods (RF, XGB, SEM, VEM) consistently outperform Logistic Regression across all datasets, achieving AURGR values between 0.5181–0.6780 compared to LR's 0.3121–0.4619, while Neural Network demonstrate intermediate performance (0.5242 and 0.5481). Under Gaussian noise, this hierarchy reverses completely: Logistic Regression achieves near-perfect robustness (0.9668–0.9815) and Neural Network show strong robustness scores (0.7608 and 0.8152), while other models remain vulnerable around 0.5070–0.5180.
- The contrasting behavior between perturbation types reveals important differences in how model architectures handle data corruption. Based on our results, Logistic Regression and Neural Network tend to maintain high robustness under Gaussian noise but experience greater degradation under Swapping perturbations. This can occur because:
  (1) Gaussian noise increases variance but preserves expected values and the general structure of feature distributions. Linear models and shallow Neural Networks can tolerate this and maintain a stable signal-to-noise ratio.
  (2) In contrast, Swapping fundamentally disrupts the learned associations between features and outcomes. This breaks the specific patterns encoded in model parameters, leading to more substantial performance degradation that cannot be readily adapted to at inference time.
  Conversely, tree-based models display the opposite trend. Their hierarchical split-based decision rules can be disrupted by Gaussian noise, as it alters the feature distributions that determine optimal split points, leading to noticeable performance drops—though ensemble averaging provides some mitigation. However, these models demonstrate better resilience to Swapping perturbations. Because trees rely on independent feature-specific thresholds at each split rather than global correlation patterns, they can maintain reasonable performance even when feature relationships are disrupted, as their local branching logic adapts to the available data structure. Importantly, the actual robustness patterns depend on perturbation severity, swapping methodology, and specific model configurations.

Table 5
AURGE comparison between datasets for each model.

Model	New York HMDA	California HMDA	Employee data
LR	0.9661	0.9644	0.9100
RF	0.9454	0.9547	0.8943
XGB	0.9416	0.9350	0.9042
SEM	0.9471	0.9406	0.9045
VEM	0.9467	0.9486	0.9002
NN	0.9414	0.9381	–
Random	1.0	1.0	1.0

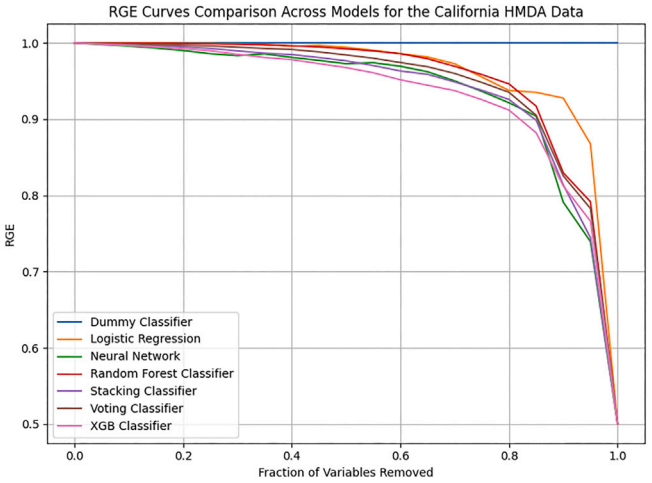


Fig. 9. Rank Graduation Explainability (RGE) Curves of all models for the California HMDA dataset. The figure indicates that about 40% of the variables can be removed, for all models.

4.3. AURGE results

We finally compare models in terms of AURGE. Table 5 contains results for all considered data and Fig. 9 visualize the RGE curves for the California HMDA data.

Based on the AURGE comparison table, here are the key conclusions:

- All models achieve AURGE scores below the Random baseline (1.0), which by construction of the formula achieves perfect explainability since variable removal does not affect its predictions: it is not a model, as no feature variable contributes to its output.
- Logistic Regression consistently demonstrates the highest explainability across all three datasets, achieving AURGE scores of 0.9661 (New York HMDA), 0.9644 (California HMDA) and 0.9100 (Synthetic Employee data). This superior performance aligns with the theoretical expectation for this metric.
- Tree-based models (RF, XGB) and ensemble methods (SEM, VEM) show comparable explainability performance, clustering in narrow ranges across datasets. XGBoost and Neural Network consistently rank among the least explainable models on all datasets.

The AURGE curve visualizations confirm that across all datasets Logistic Regression is the most explainable model, and these visualizations provide clearer differentiation between model performances compared to previous metrics graphs.

To complement the analysis, we also compare RGE scores with Shapley values for HMDA datasets. For computational efficiency and clarity of presentation, we limit this comparison to one representative model, XGBoost. We employ Spearman's rank correlation to assess the agreement between the two methods. Tables 6 and 7 present the top-ranked features obtained from the application of RGE and Shapley, respectively, along with Spearman's correlation.

Table 6
RGE and SHAP comparison for XGBoost model on New York HMDA.

Variable	RGE	Shapley	RGE rank	Shapley rank
loan_purpose	0.1266	0.5472	1	1
applicant_income	0.0852	0.3822	2	2
loan_amount	0.0756	0.2263	3	3
applicant_race_1	0.0231	0.1067	4	5
hud_median_family_income	0.0163	0.1334	5	4
minority_population	0.0077	0.0610	6	9
census_tract_number	0.0074	0.0551	7	10
Spearman's rank correlation: $\rho = 0.853$ $(p < 0.0001)$				

Table 7
RGE and SHAP comparison for XGBoost model on California HMDA.

Variable	RGE	Shapley	RGE rank	Shapley rank
loan_purpose	0.1679	0.4331	1	1
loan_amount	0.1488	0.3631	2	2
applicant_income	0.1382	0.3571	3	3
hud_median_family_income	0.0192	0.0862	4	4
applicant_race_1	0.0162	0.0667	5	6
tract_to_msamd_income	0.0150	0.0748	6	5
lien_status	0.0127	0.0182	7	18
Spearman's rank correlation: $\rho = 0.878$ $(p < 0.0001)$				

The comparison between RGE scores and Shapley values reveals strong alignment in feature importance rankings across both HMDA datasets. The Spearman's rank correlation coefficients of $\rho = 0.853$ for New York data and $\rho = 0.878$ for California data, both with $p < 0.0001$, indicate substantial agreement between the two methods. This concordance is particularly evident for the top-ranked features. Both methods identify loan_purpose, applicant_income, and loan_amount as the three most important variables across datasets, though their relative ordering slightly. The California dataset shows even stronger agreement, with the top four features (including hud_median_family_income) ranked identically by both methods. Some divergence emerges in middle-to-lower importance rankings. For example, in California, lien_status ranks seventh by RGE but eighteenth by Shapley values. These discrepancies reflect fundamental differences in methodology: RGE measures performance degradation upon feature removal through perturbation, while Shapley values calculate marginal contribution based on coalitional game theory. Features with complex interactions may be valued differently by these approaches.

Fig. 10 illustrates the comparison between RGE and Shapley for the New York HMDA (due to space constraints, only one representative visualization is included).

The visualization shows that while both methods identify the same dominant features, they exhibit greater spread for lower importance features, reflecting the different scales and aggregation methods inherent to each approach. The strong positive correlations observed across both datasets provide preliminary evidence that RGE captures similar feature importance patterns to Shapley values. However, the differences in middle-to-lower ranked features suggest that further comparative analysis across additional datasets and model architectures would be necessary to fully characterize the relationship between RGE and established explainability metrics, and to determine under what conditions they converge or diverge in their assessments.

4.4. Integration

We now calculate the integrated compliance metric directly from the original RGA, RGE, and RGR values that are used to construct the curves, without any preprocessing or transformation. The raw values

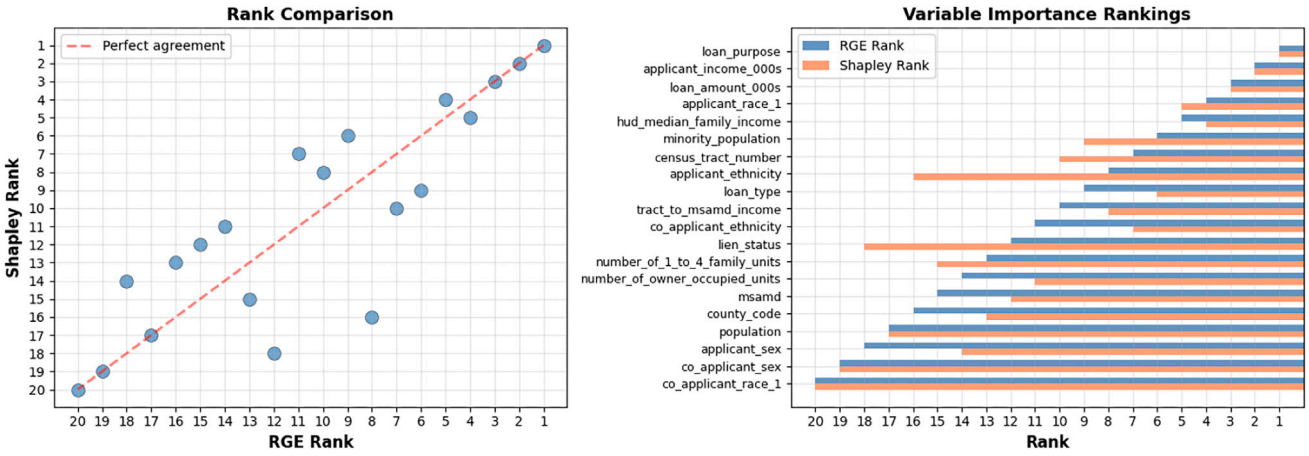


Fig. 10. Rank Graduation Explainability (RGE) and Shapley values comparison for the Gradient Boosting model on the New York HMDA. The figures indicate a strong agreement between the two variable importance methods, RGE and Shapley.

Table 8
Compliance score comparison across models using raw RGA, RGR, and RGE values for the California HMDA dataset (Swapping).

Model	Arithmetic mean	Geometric mean (1/3)	Quadratic mean (RMS)	TOPSIS
LR	0.59695	0.47965	0.67482	0.36325
RF	0.63218	0.53011	0.69337	0.40138
XGB	0.62893	0.53042	0.68661	0.40543
SEM	0.63151	0.53229	0.68968	0.40590
VEM	0.63217	0.53141	0.69202	0.40413
NN	0.62639	0.52752	0.68482	0.40140
Random	0.76681	0.61145	0.84059	0.45074

Table 9
Compliance score comparison across models using raw RGA, RGR, and RGE values for the California HMDA dataset (Gaussian noise).

Model	Arithmetic mean	Geometric mean (1/3)	Quadratic mean (RMS)	TOPSIS
LR	0.76303	0.63247	0.82379	0.45528
RF	0.62348	0.52555	0.68259	0.38863
XGB	0.61731	0.52230	0.67402	0.39080
SEM	0.61940	0.52360	0.67677	0.39091
VEM	0.62211	0.52525	0.68036	0.39058
NN	0.71322	0.60367	0.76478	0.44423
Random	0.76681	0.61145	0.84059	0.42454

from each metric vector are used as input to calculate the scores using arithmetic, cubic geometric, root mean square mean and TOPSIS methods.

For lack of space, we present the results only for the California dataset. While Tables 8 and 9 reports the results of the comparison of all models, in terms of all described mean scores and TOPSIS methodology, Fig. 11 represents all three curves for each model. Figs. 12–14 represent the distribution of the resulting values for the three considered means, using Swapping perturbations.

The main findings from Tables 8 and 9 and from Figs. 11–14 can be summarized as follows.

- Under Swapping perturbations (Table 8), ensemble methods demonstrate superior compliance:
  - The Stacked Ensemble Model (SEM) achieves the highest TOPSIS score (0.40590) and Geometric Mean (0.53229), indicating balanced performance across all three dimensions.
  - Random Forest demonstrates strong overall performance, achieving the highest Arithmetic Mean (0.63218) and Quadratic Mean (0.69337), but with a relatively low TOPSIS score (0.40138).
  - Logistic Regression ranks the lowest across all aggregation methods, reflecting its vulnerability to feature relationship disruption by Swapping perturbations.
  - Notably, all models usually cluster in a close range, suggesting comparable overall compliance under Swapping perturbations.
- Under Gaussian noise perturbations (Table 9), the performance hierarchy shifts dramatically:
  - Logistic Regression emerges as the top performer with the highest TOPSIS score and all mean-based metrics, demonstrating its resilience to additive noise that preserves linear relationships.
  - The Neural Network achieves the second-highest compliance, suggesting that its learned representations can tolerate distributional shifts from Gaussian perturbations.
  - All the other models show a degraded performance compared to swapping scenarios, indicating their sensitivity to threshold-disrupting noise.
- Figs. 11–14, for Swapping perturbations, show that the arithmetic mean is relatively right-skewed and bell-shaped distributions across all six models. The geometric mean exhibits right-skewed distributions with longer tails, while the RMS values display distinctive bimodal distributions with two unclear peaks, indicating the presence of two distinct subgroups or performance regimes in the model predictions.

These contrasting responses to different perturbation types are consistent across all analyzed datasets. However, these contradictory patterns in compliance scores—where different models excel under different perturbation regimes—indicate that further research is needed to develop unified model selection frameworks that can account for multiple degradation scenarios. Overall, applying multiple aggregation metrics provides a nuanced perspective on model compliance. While Arithmetic Mean captures general performance, Geometric Mean emphasizes balanced improvements across dimensions, RMS highlights variance sensitivity, and TOPSIS provides distance-based ranking. This approach can enable informed candidate selection for credit lending applications where accuracy, explainability, and robustness must be simultaneously optimized under realistic perturbation scenarios.

To further validate the sensitivity and interpretability of our methodology, we now demonstrate that unequal weight assignments produce rankings that directly correspond to subjective criteria. For

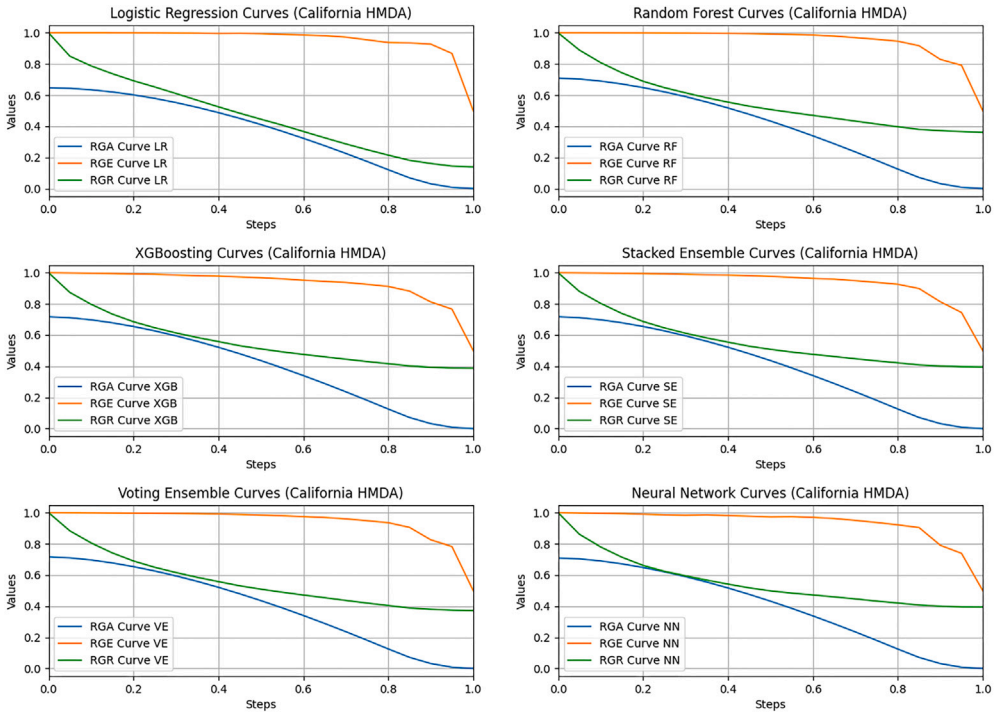


Fig. 11. Rank Graduation Accuracy (RGA), Rank Graduation Explainability (RGE) and Rank Graduation Robustness (RGR) for the California HMDA dataset (Swapping). The three curves are the basis of the unified metric.

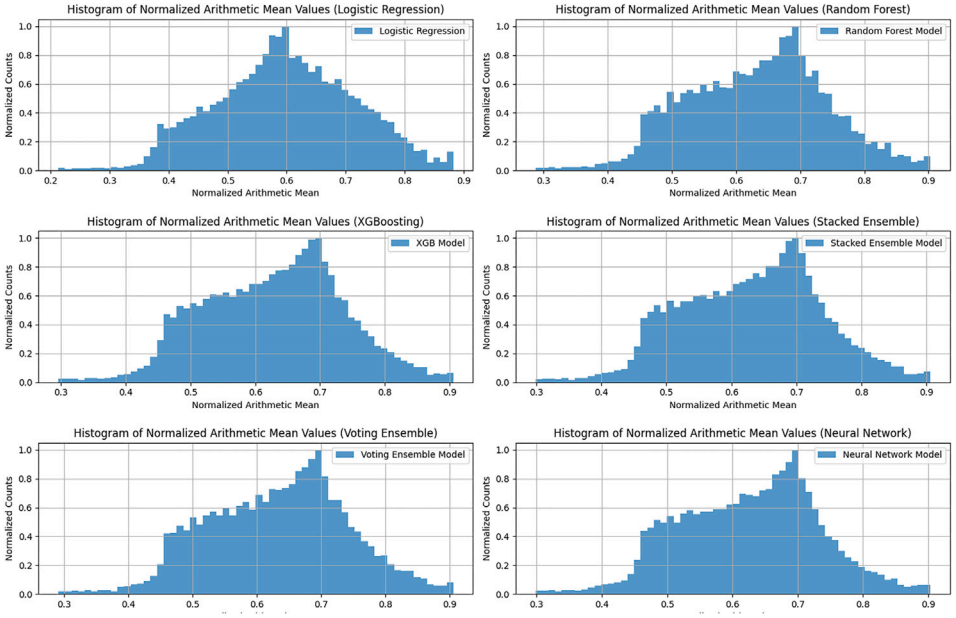


Fig. 12. Arithmetic Mean Histograms for the California HMDA dataset (Swapping). The Figures indicate the integrated performance behavior of all models, using the arithmetic mean.

simplicity, we show this on one representative dataset and for one integration method. When we assign a dominant weight of 0.9 to a specific dimension (accuracy, robustness, or explainability) and minimal weights of 0.05 to each to the remaining two dimensions, the resulting ranking should theoretically align with the raw ranking of the emphasized criterion. The following table presents the results for the New York HMDA dataset under three weighting scenarios. The weight configurations are as follows: Accuracy Focus (RGA: 0.9, RGR: 0.05, RGE: 0.05), Robustness Focus (RGA: 0.05, RGR: 0.9, RGE: 0.05) and Explainability Focus (RGA: 0.05, RGR: 0.05, RGE: 0.9) (see Table 10).

The results confirm that the framework correctly responds to weight adjustments, with model rankings shifting to reflect the emphasized dimension. This validation demonstrates that weighted aggregation

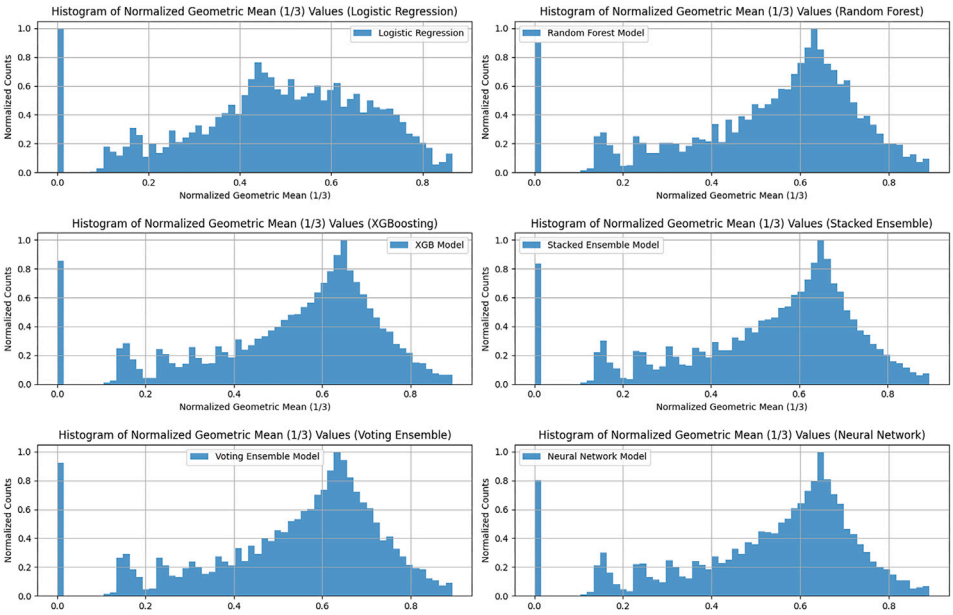


Fig. 13. Geometric Mean Histograms for the California HMDA dataset (Swapping) The Figures indicate the integrated performance behavior of all models, using the geometric mean.

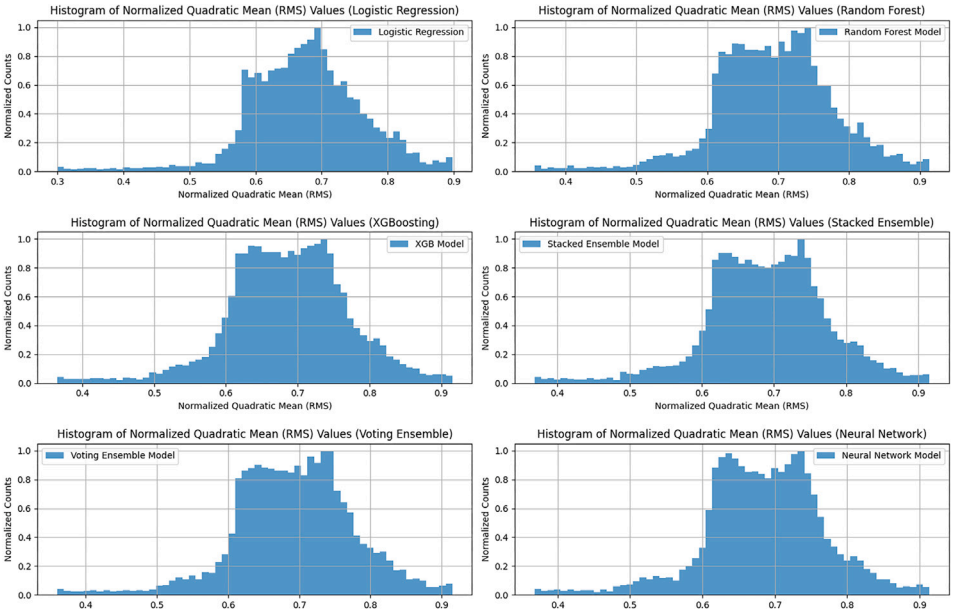


Fig. 14. Quadratic mean Histograms for the California HMDA dataset (Swapping) The Figures indicate the integrated performance behavior of all models, using the quadratic mean.

can be naturally integrated into the compliance framework in future applications, enabling organizations to customize model selection based on their specific regulatory requirements, risk appetite or operational priorities.

5. Conclusions

This work has addressed a critical gap in artificial intelligence governance by developing a unified framework for measuring S.A.F.E-AI system trustworthiness across multiple risk dimensions.

The primary contribution of this research is the introduction of three novel metrics:

- Area Under the Rank Graduation Accuracy Curve (AURGA): Evaluates model performance consistency measuring accuracy degradation across ranked prediction segments. The metric captures how accuracy changes when progressively removing contributions from the highest- to lowest-ranked predictions.
- Area Under the Rank Graduation Robustness Curve (AURGR): Assesses model stability under data perturbations using both Swapping and Gaussian noise approaches.
- Area Under the Rank Graduation Explainability Curve (AURGE): Evaluates model transparency by systematically removing features and measuring the impact on explanatory power using a greedy approach that captures how rapidly explainability is lost.

Table 10
Compliance score comparison with raw RGA, RGR (Swapping perturbations), and RGE values using TOPSIS approach for New York HMDA dataset with different focus scenarios.

Model	Accuracy focus			Robustness focus			Explainability focus		
	AURGA	TOPSIS	Rank	AURGR	TOPSIS	Rank	AURGE	TOPSIS	Rank
LR	0.3837	0.38136	6	0.4484	0.42620	7	0.9661	0.19093	2
RF	0.4121	0.41022	4	0.5181	0.49857	6	0.9454	0.18212	5
XGB	0.4133	0.41148	3	0.5372	0.51876	2	0.9416	0.18049	6
SEM	0.4137	0.41187	2	0.5279	0.50918	4	0.9471	0.18318	3
VEM	0.4138	0.41190	1	0.5289	0.50991	3	0.9467	0.18298	4
NN	0.4100	0.40812	5	0.5242	0.50584	5	0.9414	0.18006	7
Random	0.3024	0.30114	7	1.000	0.93164	1	1.000	0.21793	1

We demonstrated that model selection for compliance scoring applications requires a nuanced, multi-faceted evaluation that accounts for both absolute performance and relative stability. The comparative analysis across multiple datasets and evaluation methodologies reveals that compliance performance fundamentally depends on perturbation types. Linear models exhibit resilience to Gaussian noise but are more vulnerable to Swapping perturbations, whereas tree-based models display the opposite trend. This perturbation-dependent behavior indicates that no single architecture achieves universal superiority across all compliance dimensions; thus, the optimal model choice must consider the dominant data quality degradation modes in the target deployment environment.

The proposed Compliance Score framework, which integrates multiple aggregation methods (Arithmetic Mean, Geometric Mean, RMS, and TOPSIS), offers an evaluation tool adaptable to diverse organizational priorities through weighted schemes. The strong correlation between RGE scores and Shapley values supports the validity of the perturbation-based explainability approach, although observed discrepancies for lower-ranked features suggest the need for further investigation.

From an ethical and regulatory viewpoint, the unified metric provides a clear tool that can be employed by auditors and regulators to assess the compliance of any AI application, and to highlight where that compliance is to be improved (e.g. in security, accuracy, explainability, and also in fairness, considering the explainability of protected variables). The same auditors and regulators could also give different weights to the four dimensions, in line with the evolving policies and their implementations, as described for example, in European Commission, 2023; National Institute for Standards and Technology, 2023).

Future research should address the limitations of the present work.

First of all, our proposal should be compared with methods that integrate individual existing metrics, such as Nguyen and Do (2025), in real case scenarios, as we did for the credit data in this paper. Second, the proposal should be extended beyond tabular data, to consider, for example, image processing and natural language processing and large language models, as in Liang et al. (2022).

Another line of research that could address the limitations of the current work should concern investigation into how the Compliance Score framework can be adapted to meet evolving regulatory requirements, including the integration of fairness metrics beyond the presented measures, represents another critical research direction. This includes exploring how the framework can accommodate emerging AI governance standards and audit requirements in financial services.

Further research should focus on the challenge of reconciling conflicting model recommendations when different perturbation scenarios favor different architectures. Dynamic weighting schemes that adapt to specific regulatory requirements or business contexts represent a promising direction, potentially incorporating stakeholder preferences through interactive multi-criteria decision-making approaches. Establishing standardized benchmarking protocols and extending the framework to continuous outcome variables would broaden its applicability.

We finally remark that, while in the present paper we have, for lack of space, focused our experimental work on a credit lending database, similar analysis can be conducted for classification problems arising in other fields, such as health care and manufacturing.

CRediT authorship contribution statement

Paolo Giudici: Conceptualization, Methodology, Project administration, Resources, Supervision, Validation, Writing – review & editing. **Vasily Kolesnikov:** Data curation, Formal analysis, Investigation, Software, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is a the result of a close collaboration between the two authors. However, V.K. carried out the modeling, computational and paper drafting work, whereas P.G. conceptualized and supervised the work, and revised the draft. The authors thank the colleagues at the SAFE_AI laboratory of the University of Pavia,. for useful discussions. They also thanlk two anonymous reviewers and the editors for their constructive comments, which have helped improving the paper.

Data availability

Data will be made available on request.

References

Babaei, G., Giudici, P., & Raffinetti, E. (2025). A rank graduation box for SAFE AI. *Expert Systems with Applications*, *259*, Article 125239.

Bernardelli, A. E., & Giudici, P. (2025). AI risk management: A bibliometric analysis. *Risks*, *13*(7), 131.

Cohen, J., Rosenfeld, E., & Kolter, J. Z. (2019). Certified adversarial robustness via randomized smoothing. arXiv:1902.02918.

Doshi-Velez, F., & Kim, B. (2017). Towards a rigorous science of interpretable machine learning. arXiv.1702.08608.

European Commission (2020). *White paper on artificial intelligence: A European approach to excellence and trust*. Brussels: Report.

European Commission (2023). *The EU artificial intelligence act*. Brussels: Legislative act.

Ferrer, L., et al. (2024). Enhancing cybersecurity resilience through advanced red-teaming. *Computers & Security*, *145*, Article 102805.

Giudici, P., & Raffinetti, E. (2024). RGA: A unified measure of predictive accuracy. *Advances in Data Analysis and Classification*, *19*, 67–93.

Gneiting, T., & Katzfuss, M. (2014). Probabilistic forecasting. *Annual Review of Statistics and its Applications*, *1*(1), 125–151.

Guidotti, R., Monreale, A., Ruggieri, S., Turini, F., Giannotti, F., & Pedreschi, D. (2018). A survey of methods for explaining black box models. *ACM Computing Surveys*, *51*(5), 1–42.

Hendrycks, D., & Dietterich, T. (2019). Benchmarking neural network robustness to common corruptions and perturbations. In *International conference on learning representations*.

Hurlin, C., Pérignon, C., & Saurin, S. (2024). The fairness of credit scoring models. *Management Science*.

Hwang, C.-L., & Yoon, K. (1981). *Lecture notes in economics and mathematical systems: vol. 186, Multiple attribute decision making: Methods and applications—A state-of-the-art survey*. Springer.

International Organization for Standardization and International Electrotechnical Commission (2023). ISO/IEC 23894:2023 – Information technology – Artificial intelligence – Guidance on risk management. Standard. First edition; aligned with ISO 31000:2018 for AI risk management.

Koshevoy, G., & Mosler, K. (1996). The Lorenz zonoid of a multivariate distribution. *Journal of the American Statistical Association*, *91*(434), 873–882.

Liang, P., Bommasani, R., Lee, T., Tsipras, D., Soylu, D., Yasunaga, M., Zhang, Y., Narayanan, D., Wu, Y., Kumar, A., Newman, B., Yuan, B., Yan, B., Zhang, C., Cosgrove, C., Manning, C. D., Ré, C., Acosta-Navas, D., Hudson, D. A., ... Koreeda, Y. (2022). Holistic evaluation of language models. arXiv preprint arXiv:2211.09110. Version v2 published 1 Oct 2023; formally in TMLR Aug 2023.

Lorenz, O. M. (1905). *Methods of measuring the concentration of wealth*: *vol. 9*, (pp. 209–219). Publications of the American Statistical Association, 70.

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *vol. 30*, In *Advances in neural information processing systems*.

Molnar, C. (2019). Interpretable machine learning: A guide for making black box models explainable.

National Institute for Standards and Technology (2023). Artificial intelligence risk management framework.

National Institute for Standards and Technology (2025). Adversarial machine learning: A taxonomy and terminology of attacks and mitigations.

Nguyen, L. P. T., & Do, H. T. (2025). RAISE: A unified framework for responsible AI scoring and evaluation. arXiv preprint arXiv:2510.18559. Preprint; submitted October 2025.

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). Why should I trust you? Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 1135–1144).

Rousseeuw, P., & Leroy, A. (1987). Robust regression & outlier detection. *Journal of Educational Statistics*, *13*, 358–364.

Terven, J., Cordova-Esparza, D.-M., Romero-González, J.-A., et al. (2025). A comprehensive survey of loss functions and metrics in deep learning. *Artificial Intelligence Review*, *58*, 195.

United Nations High-Level Advisory Body on Artificial Intelligence (2024). Governing AI for humanity: final report.

Zhao, W., Alwidian, S., & Mahmoud, Q. H. (2022). Adversarial training methods for deep learning: a systematic review. *Algorithms*, *15*(8), 283.